

Martin Luther University of Halle-Wittenberg
Department of Economics
Chair of Econometrics

Econometrics II
Preliminaries and course overview

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What is econometrics?

"Intermediate between mathematics, statistics, and economics, we find a new discipline which, for lack of a better name, may be called econometrics. Econometrics has as its aim to subject abstract laws of theoretical political economy or 'pure' economics to experimental and numerical verification, and thus to turn pure economics, as far as possible, into a science in the strict sense of the word" (Frisch 1971, p. 386, first published 1926).



Ragnar Frisch
(1895 - 1973)

What is econometrics?

- “Econometrics is the field of economics that concerns itself with the application of mathematical statistics and the tools of statistical inference to the empirical measurement of relationships postulated by economic theory” (Greene 2011, pp. 41-42).
- “Econometrics is the branch of economics that provides the tools to estimate the magnitude of economic relationships” (Murray 2006, S. 1).
- Econometrics is the interaction of economic theory, observed data, and statistical methods.

Econometrics is economic data science

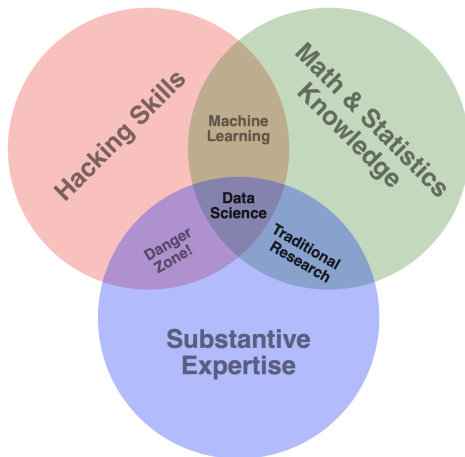


Figure: The Data Science Venn Diagram ([Conway 2010](#))

General goals

At the end of the course you will be able ...

- to apply empirical methods for estimating linear and nonlinear regression models.
- to critically evaluate empirical research.
- to use computer software for regression analyses.

Organization

- **Course material on Stud.IP:** lecture notes (slides) and material for exercise will be available via Stud.IP. Supplemental readings will be given in class and on slides.
- **Course meeting times:** see syllabus
- **Office hours:** Please register for the office hours by email at least one week in advance.

Christoph Wunder, Mon. 12:30 - 13:30

Katharina Kolb, by appointment

Organization

Final exam

- The final examination will take place in the form of a **home assignment**. The home assignment will ask students to apply econometric methods using software.
- Students need to register in person by adding their name to the registration list at the secretary's office.
- **Software:** R or Stata
- Students have **four weeks** to complete the assignment.
- Students must submit an **independently written homework**. Joint writing (co-authoring) is not allowed.

Statement of expectations

In class: You are expected ...

- to actively engage in the material covered during class.
- to participate in interactive learning.

Out of class: You are expected ...

- to complete the supplemental readings in a timely fashion.
- to continually revise the content of lectures and exercises.

Prerequisites

- Interest in the subject
- Willingness to commit time and intellectual effort
- A fundamental understanding of algebra, calculus, statistics, probability, and econometrics (Econometrics I, WIW.06756).

Some advice for learning

- Read (recommended) chapters in textbooks.
- Re-work the slides and exercise.
- Work together in small learning groups. Use collaboration platforms (e.g. GitHub).
- Write short programs using statistical software packages. It will be helpful if you have [R/RStudio](#) and [Stan](#) installed on your computer. However, other software is possible (e.g. Stata).

Course outline

Week	Topic
49/2025	Maximum likelihood estimation
50/2025	Bayesian estimation
51/2025	Binary choice models
2/2026	Panel data models
3/2026	Panel data models
4/2026	Panel data models
5/2026	Times series models
6/2026	Times series models

References

- Frisch, R. (1971). On a problem in pure economics. In Chipman, J. S., Hurwicz, L., Richter, M. K., and Sonnenschein, H. F., editors, *Preferences, Utility, and Demand. A Minnesota Symposium*, pages 386–423. Harcourt Brace Jovanovich, Inc., New York.
- Greene, W. H. (2011). *Econometric Analysis*. Prentice Hall International, Upper Saddle River, New Jersey.
- Murray, M. P. (2006). *Econometrics: A Modern Introduction*. Pearson Addison Wesley, Boston.